## PARTICLE SIZE DISTRIBUTION
## BS 1377 2:1990: 9.2/9.3/9.4 & 9.5

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Client:** | SDHUD | | | | | **Date submitted:** | 09-Jun-26 |
| **Project/Site Name:** | KIRIAINI AHP | | | | | **Date tested:** | 11-Jun-26 |
| **Sample ID:** | BH1 | **Sample No.:** | - | **Sample depth (M):** | 0.0 - 25.0 | **Sampled and Submitted by:** | Cransfield |

| **Sample preparation details** | | |
|---|---|---|
| **Initial quartered and air dried mass** | **(gm)** | 328 |
| **Washed and oven dried mass** | **(gm)** | 176.4 |
| **Fine mass** | **(gm)** | 151.6 |
| **Fines percent** | **(%)** | 46.2 |

| **Moisture content at sample preparations** | | |
|---|---|---|
| **Wet weight of sample** | **(gm)** | 186.5 |
| **Dry weight of sample** | **(gm)** | 149.0 |
| **Wet weight of water** | **(gm)** | 37.5 |
| **Moisture content** | **(%)** | 25.2 |

| **Soil Classification** | | | | | |
|---|---|---|---|---|---|
| **GRAVEL (%):** | 22.8 | **CLAY/SILT (%):** | 71.9 | **USCS** | GRAVELLY ELASTIC SILT(MH) |
| **SAND (%):** | 5.3 | | | **AASHTO** | A-7-5 |

**Wet & Dry Sieve Analysis to BS 1377 2:1990: 9.2/9.3/9.4**

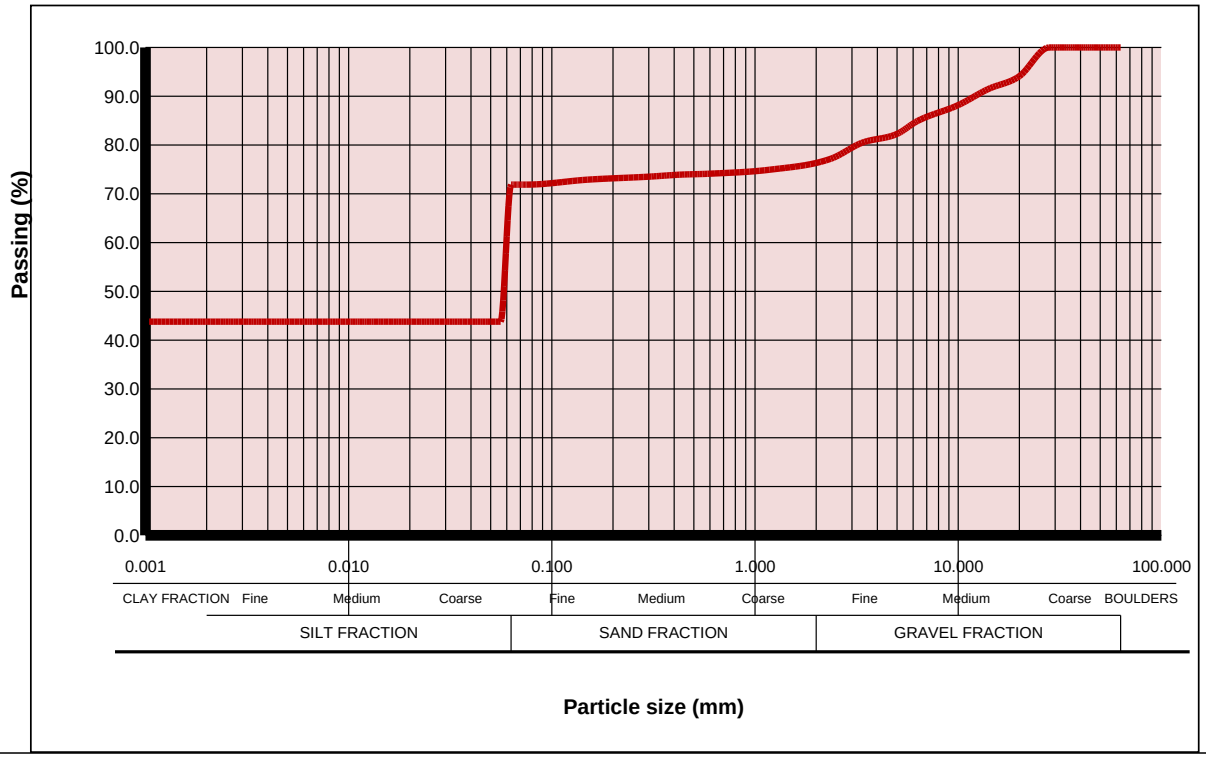
| Sieve size (mm) | Retained mass (gm) | % Retained (%) | Cumulative passing percentage (%) |
|---|---|---|---|
| **63** | 0 | 0.0 | 100.0 |
| **50** | 0 | 0.0 | 100.0 |
| **37.5** | 0 | 0.0 | 100.0 |
| **28** | 0 | 0.0 | 100.0 |
| **20** | 14.5 | 5.9 | 94.1 |
| **14** | 6.5 | 2.6 | 91.4 |
| **10** | 8.0 | 3.3 | 88.2 |
| **6.3** | 8.0 | 3.3 | 84.9 |
| **5** | 6.5 | 2.6 | 82.3 |
| **3.35** | 4.5 | 1.8 | 80.4 |
| **2.36** | 8.0 | 3.3 | 77.2 |
| **1.18** | 5.5 | 2.2 | 74.9 |
| **0.6** | 2.0 | 0.8 | 74.1 |
| **0.425** | 0.5 | 0.2 | 73.9 |
| **0.3** | 1.0 | 0.4 | 73.5 |
| **0.15** | 1.5 | 0.6 | 72.9 |
| **0.075** | 2.5 | 1.0 | 71.9 |
| **0.063** | 0.0 | 0.0 | 71.9 |
| **<0.063** | 176.4 | 71.9 | - |
| **TOTAL** | **245.4** | - | - |

**Hydrometer Analysis to BS 1377-2:1990:9.5**

| | | | |
|---|---|---|---|
| **Dry Weight** | **(g)** 176.4 | **Hydrometer Type** | 152H |
| **S.G (Mg/m³)** | 2.65 | **Zero Correction factor** | 0 |
| **Temperature (°C)** | 20 | **Meniscus correction (g/L)** | 0.1 |
| **K Factor** | 1.358 | **Temperature Correction Factor** | 0 |

| Time, min | Actual rh | Adjusted HR(Rh) | Composite correction | Corrected HR | Effective depth (cm) | Diameter; D K√(L/T) | % Fines in Suspension | % Fines based on whole sample |
|---|---|---|---|---|---|---|---|---|
| 0.25 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0792 | 43.8 | 43.8 |
| 0.50 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0560 | 43.8 | 43.8 |
| 1 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0396 | 43.8 | 43.8 |
| 2 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0280 | 43.8 | 43.8 |
| 4 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0198 | 43.8 | 43.8 |
| 8 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0140 | 43.8 | 43.8 |
| 15 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0102 | 43.8 | 43.8 |
| 30 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0072 | 43.8 | 43.8 |
| 60 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0051 | 43.8 | 43.8 |
| 120 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0036 | 43.8 | 43.8 |
| 240 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0026 | 43.8 | 43.8 |
| 480 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0018 | 43.8 | 43.8 |
| 1440 | 48 | 48.0 | 0.10 | 48.1 | 8.41 | 0.0010 | 43.8 | 43.8 |

### PARTICLE SIZE DISTRIBUTION GRAPH




| | | | | |
|---|---|---|---|---|
| **Tested by:** | JILLO | **Date reported:** | 13-Jun-26 | Checked by: _____ |